

# Alexander Bernstein

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## Education

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### University of California, Santa Barbara

*Ph.D. in Statistics*

Santa Barbara, CA

*Sept. 2025*

Dissertation: “Long-Only Minimum Variance Portfolio Composition for Factor Models”

### Washington University in St. Louis

*M.Sc. in Systems Science and Applied Mathematics*

St. Louis, MO

*Dec. 2016*

*B.A. in Mathematics and Economics (Cum Laude)*

*May 2014*

## Technical Skills

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**Quantitative Methods:** Convex & Portfolio Optimization, Factor Models, Asset Pricing (options & bonds), Black-Litterman, Risk Modeling, Time Series Analysis, Stochastic Control, Reinforcement Learning, Asymptotic Theory.

**Programming & Tools:** Python (NumPy, SciPy, Pandas, Scikit-Learn, PyTorch, CVXPY), R, MATLAB, SQL, C,  $\text{\LaTeX}$ , Git, Linux, AWS.

## Selected Projects

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*Factor-Model Portfolio Pipeline:* Spec-driven **Python** pipeline for portfolio construction from universe of S&P 500 assets. **PCA** factor models with pluggable covariance estimators (**POET** thresholding, eigenvector shrinkage, GARCH) and return estimators (**Kalman**/EKF filters, robust Student-t, Black-Litterman, L1-adaptive control); constrained mean-variance optimization via **CVXPY** (CLARABEL); weekly walk-forward backtest over a 260-week rolling window (10 PCA factors) on the S&P 500, with a live rebalancer.

*Quantitative Options Pricing:*  Implemented **Monte Carlo** (Euler/Milstein) and **finite-difference** (incl. Crank-Nicolson) solvers for the **Heston** and **CEV** models; built **Gaussian Process** (Matérn) and **PyTorch** neural-network surrogates for fast pricing and **Greeks** via autograd.

*Conditional Covariance Estimation:*  Built a **Covariance Regression** model for covariate-adjusted covariance of idiosyncratic equity returns — a parsimonious alternative to **MGARCH** that improves stability and cuts parameter count in high dimensions.

*Mean Field Game Reinforcement Learning:*  Implemented a **Q-Learning** solver for the major-minor cyber-security mean field game (Carmona & Delarue, 2018), learning optimal defense policies against an evolving population distribution  $\mu$ .

## Research

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### Publications:

Banerjee, T., **Bernstein, A.**, Feinstein, Z. (2025). “Dynamic clearing and contagion in financial networks.” *European Journal of Operational Research* 321(2), 664–675.

**Bernstein, A.**, Shkolnik, A. (2025). “Asymptotics of Quadratic Forms on a Simplex.” *In preparation*.

**Selected Talks:** INFORMS Annual Meeting (2023); SIAM Conf. on Applied & Computational Discrete Algorithms (2021, poster); CDAR Risk Seminar, UC Berkeley (2020).

## Experience

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### University of California, Santa Barbara

*Teaching Assistant & Graduate Student Mentor*

Santa Barbara, CA

*Sept. 2017 – June 2025*

Taught statistical theory, stochastic processes, regression, time series, financial mathematics, and risk theory; mentored undergraduate research in financial mathematics and optimization.

### Epic Systems

*Technical Services Engineer*

Madison, WI

*Sept. 2014 – Sept. 2015*

Tuned the Epic EHR codebase for large institutional clients; developed fixes for high-availability and regulatory compliance.

### Prozess Technologie

*Intern*

St. Louis, MO

*May 2013 – May 2014*

Built computational simulations to validate laboratory equipment and calibrated pharmaceutical manufacturing hardware to industry standards.